

Qualification Pack



Broadband Technician

QP Code: TEL/Q0102

Version: 5.0

NSQF Level: 4

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TEL/Q0102: Broadband Technician

Brief Job Description

A Broadband Technician installs, configures, and tests CPE (modems, routers, switches) for broadband access. The individual ensures seamless connectivity between CPE and end-user devices (CPU, laptop, tablets, Smart/IP TV) at customer premises and troubleshoots connectivity, cable, and equipment issues in coordination with the NOC.

Personal Attributes

The individual should have good communication, interpersonal, and teamwork skills, along with a technical mindset and problem-solving ability. The individual should be customer-focused and able to work under pressure with practical judgment.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TEL/N0111: Lay cable/system wiring and install equipment at customer premises](#)
2. [TEL/N0112: Configure customer premises equipment and establish broadband connectivity](#)
3. [TEL/N0113: Troubleshoot and rectify faults](#)
4. [TEL/N9105: Follow sustainable practices in telecom infrastructure installation](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Customer Service - Passive Infrastructure
Country	India
NSQF Level	4
Credits	15
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3114.0804

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Minimum Educational Qualification & Experience	12th grade Pass OR Completed 2nd year of the 3-year diploma after 10 (in Electronics and Communication Engineering/ Electrical Engineering/ Information Technology) OR 10th grade pass with 3 Years of experience in the optical fiber/ broadband domain OR Previous relevant Qualification of NSQF Level (3.0) with 3 Years of experience in the optical fiber/ broadband domain OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience in the optical fiber/ broadband domain
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	08/05/2025
Version	5.0
Reference code on NQR	QG-04-TL-04068-2025-V2-TSSC
NQR Version	2

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TEL/N0111: Lay cable/system wiring and install equipment at customer premises

Description

This unit covers the installation of system wiring, and setting up equipment, including Power over Ethernet (PoE)-supported devices such as IP cameras and VoIP phones, at customer premises.

Scope

The scope covers the following :

- Prepare for wiring and equipment installation
- Undertake wiring and install system hardware
- Install/replace UPS and check domestic power supply
- Clean up worksite and complete documentation

Elements and Performance Criteria

Prepare for wiring and equipment installation

To be competent, the user/individual on the job must be able to:

- PC1.** collect work requirements, tools, equipment, and materials required for installation, ensuring availability of PoE-compatible components
- PC2.** visit site/customer premises for installation, assessing feasibility for PoE-supported devices
- PC3.** analyze the installation environment and customer requirements to select the correct type of cables, connectors, and PoE injectors/switches
- PC4.** inspect indoor and outdoor cable routes to ensure they are free of electrical hazards and interference sources
- PC5.** verify that the cable running length is within permissible limits to maintain signal continuity and throughput, considering power transmission for PoE devices
- PC6.** ensure the equipment installation location is near a power point, has proper signal coverage, and supports PoE power budgets

Undertake wiring and install system hardware

To be competent, the user/individual on the job must be able to:

- PC7.** install structured wiring (interior and exterior) from PoP to customer premises, integrating PoE network elements as required
- PC8.** perform cable splicing and crimping, ensuring proper termination for both standard and PoE-enabled cables
- PC9.** perform neat wiring and clipping within customer premises, adhering to structured cabling norms
- PC10.** use appropriate connectors and terminate cables securely, ensuring PoE-supported devices receive stable power and data signals
- PC11.** perform immediate fault clearance if connectivity or power issues arise during installation
- PC12.** test cables and joints for transmission loss and signal strength, re-terminating if loss exceeds prescribed limits

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PC13. install equipment such as modems, routers, PoE switches, IP cameras, and VoIP phones, ensuring proper connectivity

PC14. configure and verify the functionality of PoE-supported devices to ensure seamless integration

PC15. explain the use of installed equipment to customers, including PoE-enabled devices

Install/replace UPS and check domestic power supply

To be competent, the user/individual on the job must be able to:

PC16. perform voltage, current, and earthing checks to ensure a stable power supply

PC17. inspect and replace UPS batteries in case of failure, ensuring uninterrupted power for network equipment

PC18. install/replace UPS as per manufacturer's instructions

PC19. route the power supply through UPS to protect network infrastructure

PC20. calculate equipment load and compare it with UPS capacity to prevent overloading

Clean up worksite and complete documentation

To be competent, the user/individual on the job must be able to:

PC21. dispose of installation waste properly and restore the worksite

PC22. record installation details, test results, and update network plans, including PoE device configurations

PC23. complete installation documents and obtain customer sign-off

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. specification of network cables, PoE devices, and network equipment

KU2. different sizes, colours, and categories of wires, including CAT5e, CAT6, CAT6A, and fiber optic cables

KU3. different types of cables (OFC, UTP, STP, Twisted Pair, etc.) and connectors (RJ-45, RJ-11, SC, LC, etc.)

KU4. structured cabling standards for conventional and PoE-supported installations

KU5. crimping, splicing, and termination techniques for different cable types, ensuring proper power delivery for PoE

KU6. measuring distances accurately to maintain permissible limits for network performance and PoE voltage drop

KU7. selection of suitable installation locations while adhering to cabling norms and ensuring sufficient power for PoE devices

KU8. functions, configurations and troubleshooting of customer premise equipment, including modems, routers, PoE switches, IP cameras, and VoIP phones

KU9. safe handling and use of diagnostic tools such as LAN testers, PoE testers, and network analyzers

KU10. usage of hand and power tools for installation, cable routing, and device mounting

KU11. immediate troubleshooting techniques during installation, including resolving PoE power issues and basic connectivity faults

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- KU12.** risks and impacts of not following structured wiring guidelines, safety procedures, and PoE power limits
- KU13.** documentation and record-keeping requirements for network infrastructure and PoE installations
- KU14.** payment processing procedures and customer communication best practices

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret installation diagrams, device manuals, and network layouts
- GS2.** communicate effectively with customers and team members regarding installation progress and technical details
- GS3.** liaise with vendors, suppliers, and customers to coordinate material procurement and service delivery
- GS4.** follow safety guidelines and regulatory compliance while working with electrical components and PoE devices
- GS5.** apply problem-solving skills to resolve immediate installation issues related to cabling, connectivity, and PoE power distribution
- GS6.** work efficiently under time constraints while maintaining quality standards
- GS7.** maintain proper documentation of network configurations, equipment specifications, and customer feedback
- GS8.** adapt to emerging industry trends, such as higher-power PoE standards (IEEE 802.3bt)
- GS9.** use digital tools and applications to log service records and update customer details
- GS10.** maintain a customer-centric approach, ensuring clarity in explaining installed equipment functionalities

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for wiring and equipment installation</i>	6	12	-	5
PC1. collect work requirements, tools, equipment, and materials required for installation, ensuring availability of PoE-compatible components	1	2	-	1
PC2. visit site/customer premises for installation, assessing feasibility for PoE-supported devices	1	2	-	1
PC3. analyze the installation environment and customer requirements to select the correct type of cables, connectors, and PoE injectors/switches	1	2	-	1
PC4. inspect indoor and outdoor cable routes to ensure they are free of electrical hazards and interference sources	1	2	-	1
PC5. verify that the cable running length is within permissible limits to maintain signal continuity and throughput, considering power transmission for PoE devices	1	2	-	0.5
PC6. ensure the equipment installation location is near a power point, has proper signal coverage, and supports PoE power budgets	1	2	-	0.5
<i>Undertake wiring and install system hardware</i>	11	22	-	8
PC7. install structured wiring (interior and exterior) from PoP to customer premises, integrating PoE network elements as required	1.5	2.5	-	1
PC8. perform cable splicing and crimping, ensuring proper termination for both standard and PoE-enabled cables	1.5	2.5	-	1
PC9. perform neat wiring and clipping within customer premises, adhering to structured cabling norms	1.5	2.5	-	1
PC10. use appropriate connectors and terminate cables securely, ensuring PoE-supported devices receive stable power and data signals	1	2.5	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. perform immediate fault clearance if connectivity or power issues arise during installation	1	2.5	-	1
PC12. test cables and joints for transmission loss and signal strength, re-terminating if loss exceeds prescribed limits	1.5	2.5	-	1
PC13. install equipment such as modems, routers, PoE switches, IP cameras, and VoIP phones, ensuring proper connectivity	1	2.5	-	1
PC14. configure and verify the functionality of PoE-supported devices to ensure seamless integration	1	2.5	-	0.5
PC15. explain the use of installed equipment to customers, including PoE-enabled devices	1	2	-	0.5
<i>Install/replace UPS and check domestic power supply</i>	10	10	-	5
PC16. perform voltage, current, and earthing checks to ensure a stable power supply	2	2	-	1
PC17. inspect and replace UPS batteries in case of failure, ensuring uninterrupted power for network equipment	2	2	-	1
PC18. install/replace UPS as per manufacturer's instructions	2	2	-	1
PC19. route the power supply through UPS to protect network infrastructure	2	2	-	1
PC20. calculate equipment load and compare it with UPS capacity to prevent overloading	2	2	-	1
<i>Clean up worksite and complete documentation</i>	3	6	-	2
PC21. dispose of installation waste properly and restore the worksite	1	2	-	1
PC22. record installation details, test results, and update network plans, including PoE device configurations	1	2	-	0.5
PC23. complete installation documents and obtain customer sign-off	1	2	-	0.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	TEL/N0111
NOS Name	Lay cable/system wiring and install equipment at customer premises
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Customer Service - Passive Infrastructure
NSQF Level	4
Credits	5
Version	6.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

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TEL/N0112: Configure customer premises equipment and establish broadband connectivity

Description

This unit covers CPE configuration, service provider gateway and end-user device connectivity, broadband integration with smart home ecosystems, basic cybersecurity implementation, and customer documentation of configurations and testing.

Scope

The scope covers the following :

- Configure CPE
- Connect CPE to the service provider gateway and end-user devices
- Record configuration settings and testing steps for the customer

Elements and Performance Criteria

Configure CPE

To be competent, the user/individual on the job must be able to:

- PC1.** connect a laptop/PC, smart/IP TV, IoT devices, and other customer devices to the CPE and establish connectivity
- PC2.** access Customer Premises Equipment (CPE) settings using default login credentials and update them for security
- PC3.** configure CPE as per base settings (IP, gateway, mask, NAT, QoS, etc.) and enable IPv6 support
- PC4.** set up VPN or Internet Lease Line (ILL) configurations as per customer requirements
- PC5.** follow basic cybersecurity settings such as strong password policies, firewalls, and MAC filtering

Connect CPE to the service provider gateway and end-user devices

To be competent, the user/individual on the job must be able to:

- PC6.** verify that all cables and connectors are properly plugged in and functional
- PC7.** ping the service provider gateway and analyze response time for troubleshooting
- PC8.** analyze connectivity test results for latency, throughput, and packet loss parameters
- PC9.** configure end-user devices to establish LAN/Wi-Fi connectivity with CPE, including SSID and security settings
- PC10.** ping CPE from the end-user device, analyze the response, and optimize network settings for stability
- PC11.** enable Quality of Service (QoS) settings to prioritize network traffic based on user needs
- PC12.** integrate broadband with smart home systems like Amazon Alexa, Google Home, or Apple HomeKit

Record configuration settings and testing steps for the customer

To be competent, the user/individual on the job must be able to:

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- PC13.** record CPE configuration settings, including network security configurations and VPN/ILL setups
- PC14.** record end-user device configuration settings, including IP allocation and firewall settings
- PC15.** document the pinging procedure and expected result parameters for troubleshooting reference
- PC16.** perform a speed test, record throughput data, and demonstrate performance as per the subscribed plan
- PC17.** brief the customer on basic troubleshooting steps/self-help techniques, including cybersecurity best practices
- PC18.** provide guidance on monitoring network activity and updating firmware for security and performance improvements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** risk and impact of not following defined procedures/work instructions issued as per SHE & OSH guidelines
- KU2.** command-line access and command prompts to execute basic networking commands
- KU3.** configuration settings of CPE (wired and wireless), including VLAN, NAT, and QoS settings
- KU4.** the basics of VPNs, Internet Lease Lines (ILL), and their role in secure network communications
- KU5.** advanced networking concepts including IPv6 addressing, subnetting, Network Address Translation (NAT) configurations, and Quality of Service (QoS) and their impact on broadband services
- KU6.** cybersecurity fundamentals, including securing home networks, firewall configurations, and threat mitigation strategies
- KU7.** connectivity options and methods for CPE and end-user devices, including advanced Wi-Fi security settings
- KU8.** Customer Premises Equipment (CPE) types, features, and integration with smart home systems
- KU9.** the features and operating requirements of test equipment for network diagnostics and performance evaluation
- KU10.** speed test methodologies, acceptable latency, jitter values, and how to demonstrate network performance to the customer
- KU11.** Level 1 and 2 diagnostics for resolving common network issues efficiently
- KU12.** how to Interpret in-built diagnostic results for remedial action and escalating as required
- KU13.** escalation matrix for reporting incidents, troubleshooting major network failures, and handling emergencies

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** interact courteously with customers while troubleshooting and providing solutions

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- GS2.** liaise with customers and vendors to address service requests and technical issues
- GS3.** work in coordination with the team to resolve complex network issues efficiently
- GS4.** communicate effectively in the local language to guide customers on self-help and troubleshooting
- GS5.** work systematically with attention to detail while configuring network settings and implementing cybersecurity measures
- GS6.** maintain accurate records of configuration settings, test results, and troubleshooting steps
- GS7.** adapt to emerging networking technologies, including smart home integrations and security advancements
- GS8.** provide clear and concise guidance to customers on best practices for network security and optimal broadband usage
- GS9.** analyze connectivity trends and identify potential network improvements to enhance customer experience
- GS10.** follow all safety protocols and ensure compliance with regulatory guidelines for network installations and troubleshooting

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Configure CPE</i>	10	15	-	5
PC1. connect a laptop/PC, smart/IP TV, IoT devices, and other customer devices to the CPE and establish connectivity	2	3	-	1
PC2. access Customer Premises Equipment (CPE) settings using default login credentials and update them for security	2	3	-	1
PC3. configure CPE as per base settings (IP, gateway, mask, NAT, QoS, etc.) and enable IPv6 support	2	3	-	1
PC4. set up VPN or Internet Lease Line (ILL) configurations as per customer requirements	2	3	-	1
PC5. follow basic cybersecurity settings such as strong password policies, firewalls, and MAC filtering	2	3	-	1
<i>Connect CPE to the service provider gateway and end-user devices</i>	14	21	-	9
PC6. verify that all cables and connectors are properly plugged in and functional	2	3	-	1.5
PC7. ping the service provider gateway and analyze response time for troubleshooting	2	3	-	1.5
PC8. analyze connectivity test results for latency, throughput, and packet loss parameters	2	3	-	1.5
PC9. configure end-user devices to establish LAN/Wi-Fi connectivity with CPE, including SSID and security settings	2	3	-	1.5
PC10. ping CPE from the end-user device, analyze the response, and optimize network settings for stability	2	3	-	1
PC11. enable Quality of Service (QoS) settings to prioritize network traffic based on user needs	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. integrate broadband with smart home systems like Amazon Alexa, Google Home, or Apple HomeKit	2	3	-	1
<i>Record configuration settings and testing steps for the customer</i>	6	14	-	6
PC13. record CPE configuration settings, including network security configurations and VPN/ILL setups	1	3	-	1
PC14. record end-user device configuration settings, including IP allocation and firewall settings	1	2	-	1
PC15. document the pinging procedure and expected result parameters for troubleshooting reference	1	2	-	1
PC16. perform a speed test, record throughput data, and demonstrate performance as per the subscribed plan	1	3	-	1
PC17. brief the customer on basic troubleshooting steps/self-help techniques, including cybersecurity best practices	1	2	-	1
PC18. provide guidance on monitoring network activity and updating firmware for security and performance improvements	1	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	TEL/N0112
NOS Name	Configure customer premises equipment and establish broadband connectivity
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Customer Service - Passive Infrastructure
NSQF Level	4
Credits	4
Version	6.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

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TEL/N0113: Troubleshoot and rectify faults

Description

This unit covers troubleshooting and resolving cable, connector, CPE, and broadband service faults using traditional and AI-based tools, including remote diagnostics, documentation, and site restoration.

Scope

The scope covers the following :

- Troubleshoot and repair cable, connector, and CPE faults
- Troubleshoot and restore broadband service
- Document work and clean the site

Elements and Performance Criteria

Troubleshoot and repair cable, connector, and CPE faults

To be competent, the user/individual on the job must be able to:

- PC1.** identify the root cause of faults, including No Service, service degradation, and intermittent connectivity issues
- PC2.** use signal level meters, OTDR, or AI-based diagnostic tools to test cabling and identify faults
- PC3.** repair or replace faulty connectors and damaged cables using industry-standard tools and techniques
- PC4.** perform re-connectorization/crimping of cable pairs or replace cables to restore connectivity
- PC5.** connect CPE to a laptop, CPU, or portable device for diagnostics and troubleshooting
- PC6.** access the CPE through an appropriate browser/software application, run diagnostic tests, and analyze logs
- PC7.** install or update CPE access software if required for troubleshooting or configuration
- PC8.** reset, reconfigure, or update firmware/software settings on CPE to resolve faults
- PC9.** utilize remote configuration and diagnostic tools to identify and troubleshoot faults without an on-site visit

Troubleshoot and restore broadband service

To be competent, the user/individual on the job must be able to:

- PC10.** diagnose and repair faults between customer equipment and the optical node using AI-assisted analysis
- PC11.** identify and rectify signal loss, interference, or line attenuation using advanced measurement tools
- PC12.** take readings at splitter points and terminated ends to determine signal integrity and continuity
- PC13.** conduct network troubleshooting, including ping tests, trace routes, and speed tests, using remote diagnostic tools
- PC14.** detect and document system, drop, and in-house signal leakage for resolution
- PC15.** assist customers in resolving issues using remote configuration tools and AI-driven troubleshooting guidance

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Document work and clean the site

To be competent, the user/individual on the job must be able to:

- PC16.** log troubleshooting steps, fault isolation procedures, and test results in the system database
- PC17.** record all repairs, replacements, and software updates performed on CPE or network components
- PC18.** restore any modifications made to the worksite and ensure customer satisfaction

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** basic principles of transmission, switching, and broadband communication systems
- KU2.** circuit board components, processors, and their role in telecommunication devices
- KU3.** various types of cables, cable pairs, and their signal transmission properties
- KU4.** cable connectors, their classifications, and the impact of improper connections
- KU5.** tools and equipment required for troubleshooting cable and connector faults
- KU6.** crimping, soldering techniques, and their application in network repairs
- KU7.** EMI/EMC interference and mitigation strategies specific to modems and broadband devices
- KU8.** use of signal level meters, OTDR, and AI-assisted fault detection tools for diagnostics
- KU9.** use of AI-based troubleshooting software for predictive maintenance and fault identification
- KU10.** remote diagnostic techniques and applications for CPE and broadband services
- KU11.** command-line configurations and software-based rectification methods for CPE
- KU12.** interpretation of signal attenuation, tiling, and interference patterns
- KU13.** risk factors and consequences of incorrect troubleshooting procedures
- KU14.** escalation matrix for reporting unresolved faults and system failures

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively with customers and team members to understand service issues and provide solutions
- GS2.** maintain systematic and accurate records of troubleshooting activities and resolutions
- GS3.** operate troubleshooting tools, test meters, and remote diagnostic software efficiently
- GS4.** adapt to evolving AI-based troubleshooting and remote configuration technologies
- GS5.** provide clear technical instructions to non-technical users/customers
- GS6.** follow safety protocols while handling cables, connectors, and electrical equipment
- GS7.** work collaboratively in a team setting, ensuring smooth resolution of service faults
- GS8.** apply critical thinking to diagnose faults and determine the best corrective measures
- GS9.** manage time effectively to handle multiple service requests promptly
- GS10.** uphold customer satisfaction by ensuring service restoration with minimal downtime

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Troubleshoot and repair cable, connector, and CPE faults</i>	18	27	-	9
PC1. identify the root cause of faults, including No Service, service degradation, and intermittent connectivity issues	2	3	-	1
PC2. use signal level meters, OTDR, or AI-based diagnostic tools to test cabling and identify faults	2	3	-	1
PC3. repair or replace faulty connectors and damaged cables using industry-standard tools and techniques	2	3	-	1
PC4. perform re-connectorization/crimping of cable pairs or replace cables to restore connectivity	2	3	-	1
PC5. connect CPE to a laptop, CPU, or portable device for diagnostics and troubleshooting	2	3	-	1
PC6. access the CPE through an appropriate browser/software application, run diagnostic tests, and analyze logs	2	3	-	1
PC7. install or update CPE access software if required for troubleshooting or configuration	2	3	-	1
PC8. reset, reconfigure, or update firmware/software settings on CPE to resolve faults	2	3	-	1
PC9. utilize remote configuration and diagnostic tools to identify and troubleshoot faults without an on-site visit	2	3	-	1
<i>Troubleshoot and restore broadband service</i>	9	16	-	8
PC10. diagnose and repair faults between customer equipment and the optical node using AI-assisted analysis	2	3	-	1.5
PC11. identify and rectify signal loss, interference, or line attenuation using advanced measurement tools	2	3	-	1.5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. take readings at splitter points and terminated ends to determine signal integrity and continuity	2	3	-	1.5
PC13. conduct network troubleshooting, including ping tests, trace routes, and speed tests, using remote diagnostic tools	1	2	-	1.5
PC14. detect and document system, drop, and in-house signal leakage for resolution	1	3	-	1
PC15. assist customers in resolving issues using remote configuration tools and AI-driven troubleshooting guidance	1	2	-	1
<i>Document work and clean the site</i>	3	7	-	3
PC16. log troubleshooting steps, fault isolation procedures, and test results in the system database	1	3	-	1
PC17. record all repairs, replacements, and software updates performed on CPE or network components	1	2	-	1
PC18. restore any modifications made to the worksite and ensure customer satisfaction	1	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	TEL/N0113
NOS Name	Troubleshoot and rectify faults
Sector	Telecom
Sub-Sector	Passive Infrastructure
Occupation	Customer Service - Passive Infrastructure
NSQF Level	4
Credits	4
Version	5.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

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TEL/N9105: Follow sustainable practices in telecom infrastructure installation

Description

This unit covers sustainable practices in telecom infrastructure installation. It includes identifying components for recycling or refurbishment, ensuring compliance with environmental standards, and promoting sustainability during installation and repair processes.

Scope

The scope covers the following :

- Segregate recyclable and refurbishable components
- Dispose of waste
- Use Energy-Efficient Methods
- Follow environmental standards and compliance guidelines
- Guide team members

Elements and Performance Criteria

Segregate recyclable and refurbishable components

To be competent, the user/individual on the job must be able to:

- PC1.** identify telecom components suitable for recycling or refurbishment
- PC2.** sort electronic and non-electronic waste based on disposal protocols
- PC3.** label and store recyclable and refurbishable components separately

Dispose of waste

To be competent, the user/individual on the job must be able to:

- PC4.** follow approved procedures for the safe disposal of hazardous and non-hazardous waste
- PC5.** coordinate with authorized e-waste recycling units or certified disposal agencies
- PC6.** handle batteries, cables, and electronic circuits safely to prevent environmental hazards

Use Energy-Efficient Methods

To be competent, the user/individual on the job must be able to:

- PC7.** select and use energy-efficient tools and equipment during telecom installations
- PC8.** apply best practices to minimize energy consumption
- PC9.** position and install telecom infrastructure to optimize efficiency and sustainability

Follow environmental standards and compliance guidelines

To be competent, the user/individual on the job must be able to:

- PC10.** adhere to national and international environmental regulations for telecom infrastructure installation
- PC11.** maintain records of waste disposal, recycling, and sustainability measures
- PC12.** assist in periodic audits to assess compliance with green standards
- PC13.** follow workplace sustainability protocols and report non-compliance

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Guide team members

To be competent, the user/individual on the job must be able to:

PC14. guide team members on sustainable telecom installation guidelines and practices

PC15. encourage environmentally responsible work habits

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the national and international environmental laws and regulations

KU2. e-waste management and recycling policies

KU3. occupational safety and health standards related to environmental practices

KU4. the types of recyclable and refurbishable telecom components

KU5. methods for reducing electronic waste through responsible procurement and reuse

KU6. advancements in eco-friendly telecom infrastructure

KU7. techniques for optimizing energy consumption in telecom operations

KU8. use of renewable energy sources in telecom infrastructure

KU9. best practices for reducing the carbon footprint of telecom installations

KU10. identification and proper disposal methods for hazardous and non-hazardous waste

KU11. the procedures for handling, storing, and transporting waste materials safely

KU12. how to collaborate with authorized agencies for effective waste collection and disposal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively with team members to ensure adherence to sustainable practices

GS2. follow written instructions for waste segregation, disposal, and energy-efficient installations

GS3. use problem-solving skills to address sustainability challenges in telecom installations

GS4. apply numerical skills to track and record environmental compliance data

GS5. organize tasks efficiently to minimize resource wastage and environmental impact

GS6. adapt to new sustainable technologies in telecom infrastructure

GS7. work collaboratively with authorized agencies to ensure proper e-waste disposal

GS8. maintain safety protocols while handling hazardous materials

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Segregate recyclable and refurbishable components</i>	6	12	-	6
PC1. identify telecom components suitable for recycling or refurbishment	2	4	-	2
PC2. sort electronic and non-electronic waste based on disposal protocols	2	4	-	2
PC3. label and store recyclable and refurbishable components separately	2	4	-	2
<i>Dispose of waste</i>	6	10	-	5
PC4. follow approved procedures for the safe disposal of hazardous and non-hazardous waste	2	4	-	2
PC5. coordinate with authorized e-waste recycling units or certified disposal agencies	2	3	-	2
PC6. handle batteries, cables, and electronic circuits safely to prevent environmental hazards	2	3	-	1
<i>Use Energy-Efficient Methods</i>	6	10	-	3
PC7. select and use energy-efficient tools and equipment during telecom installations	2	4	-	1
PC8. apply best practices to minimize energy consumption	2	3	-	1
PC9. position and install telecom infrastructure to optimize efficiency and sustainability	2	3	-	1
<i>Follow environmental standards and compliance guidelines</i>	8	12	-	4
PC10. adhere to national and international environmental regulations for telecom infrastructure installation	2	3	-	1
PC11. maintain records of waste disposal, recycling, and sustainability measures	2	3	-	1
PC12. assist in periodic audits to assess compliance with green standards	2	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. follow workplace sustainability protocols and report non-compliance	2	3	-	1
<i>Guide team members</i>	4	6	-	2
PC14. guide team members on sustainable telecom installation guidelines and practices	2	3	-	1
PC15. encourage environmentally responsible work habits	2	3	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TEL/N9105
NOS Name	Follow sustainable practices in telecom infrastructure installation
Sector	Telecom
Sub-Sector	
Occupation	Generic
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	30/04/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TEL/N0111.Lay cable/system wiring and install equipment at customer premises	30	50	-	20	100	25
TEL/N0112.Configure customer premises equipment and establish broadband connectivity	30	50	-	20	100	25
TEL/N0113.Troubleshoot and rectify faults	30	50	-	20	100	25
TEL/N9105.Follow sustainable practices in telecom infrastructure installation	30	50	0	20	100	15
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	140	230	0	80	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
TSSC	Telecom Sector Skill Council
TRAI	Telecom Regulatory Authority of India
LAN	Local Area Network
WAN	Wide Area Network
GI	Galvanized Iron
RCC	Reinforced Cement Concrete
PLB	Permanently Lubricated Duct
OTDR	Optical Time-Domain Reflectometer
OHS	Occupational Health and Safety
EPR	Earth Potential Rise
AT	Acceptance Testing
NOC	Network Operations Center
OFCs	Optical Fiber Cables
STMs	Synchronous Transport Modules
VFL	Visual Fault Locator
POPs	Points of Presence
TAT	Turn Around Time
SLAs	Service Level Agreements
KPIs	Key Performance Indicators
WDM	Wavelength Division Multiplexing
FRP	Fiber Reinforced Plastic

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components
Occupation	Occupation is a set of job roles, which perform similar/related set of functions in an industry.
Function	Function is an activity necessary for achieving the key purpose of the sector, occupation, or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of OS.
Job Role	Job role defines a unique set of functions that together form a unique employment opportunity in an organization.
OS	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the knowledge and understanding they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.

Qualification Pack

Performance Criteria	Performance Criteria are statements that together specify the standard of performance required when carrying out a task.
NOS	NOS are Occupational Standards which apply uniquely in the Indian context.
Qualifications Pack Code	Qualifications Pack Code is a unique reference code that identifies a qualifications pack.
Qualifications Pack	Qualifications Pack comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A qualifications Pack is assigned a unique qualification pack code.
Unit Code	Unit Code is a unique identifier for an Occupational Standard, which is denoted by an 'N' followed by a number.
Unit Title	Unit Title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Knowledge and Understanding	Knowledge and Understanding are statements which together specify the technical, generic, professional and organizational specific knowledge that an individual needs in order to perform to the required standard.
Organizational Context	Organizational Context includes the way the organization is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical Knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
'Core Skills or Generic Skills	Core Skills or Generic Skills are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.'